

Executive Summary – Catchment Area Population and Access

Background: On April 23rd, 2024, SCI D GIS Client requested an analysis of Catchment Area Population and Access. The following questions were submitted:

For each of the 25 catchment areas, completed for all diagnoses, as well as SCI/D only (two separate reports).

- Total number of Veterans in that catchment area (all those where their closest hub is in that catchment area)
- Total number of Veterans in each closest-hub-or-spoke for the catchment area (hub and each spoke numbers)
- Analysis for the catchment area...did the population for the entire catchment area, and for each individual spoke and the hub, have a statistically significant increase or decrease during FY19 to FY23?
- Are there catchment areas, or individual hubs and spokes, which have had the most increase or decrease in Veterans living closest? Another way to ask the question is, are there any geographic catchment areas, or individual hubs and spokes, that are significantly growing or shrinking?
- For each catchment area, and each individual hub and spoke, provide the average drive time and distance to closest hub, and closest hub-or-spoke. The questions are:
 - Do certain hubs, certain spokes, and certain overall catchment areas, have Veterans that have to drive statistically significantly further, or spend more time driving, than other catchment areas, and does this change over time (FY19-23)?
 - Is there a relationship between catchment area or hub population size and change FY19-23, and drive time/distance? In other words, do catchment areas who have a relative, significant population increase over time, have less average drive time or distance, and would the reverse be true for catchment areas and hubs with relative, significant population decreases?

This executive summary will provide a research abstract, the methodology of our work, the analysis, and a quick synopsis of all the questions posed. All of the requested analysis, population tables, and mapping products are included in the two presentations that were created for this request. They are located here:

- [Spatial Temporal Analysis of SCI/D Veteran Population \(All Dx\)](#) by Facility Territory from FY19 to FY23
- [Spatial-Temporal Analysis of SCI/D Veteran Population \(SCI Dx Only\)](#) by Facility Territory from FY19 to FY23

All Dx Abstract:

A spatial-temporal analysis was done on the VHA SCI/D Veteran Population to better understand veteran proximity to specialty care as well as population change over time for the last five years. Annual population data for 25,157 veterans was ingested into a GIS spatial software for spatial analysis. Veteran addresses were geocoded and geographic territories were established for each facility in the VHA SCID System of Care within the United States and Puerto Rico (ESRI, ArcGIS Pro Desktop V11.3). The total study area was 3,655,250 mi², the average Hub facility territory area was 146,210 mi², and the average spoke territory was 24,775mi².

The Veteran population was summarized by facility using the Closest Facility tool in the Network Analyst toolbox (ESRI, ArcGIS Pro Desktop V3.3). The 5 year average veteran population by Hub was 991 veterans, and ranged from 253 in Hampton to 2,257 in Augusta. The average Spoke population was 170 veterans, and ranged from 25 in Hawaii to 824 in Dallas. The total net change in veteran population over five years was +240 veterans, with an average annual change of 60. Seven hubs had significant change (outside ± 2 SD: Aurora +237; outside ± 1 SD: Richmond +125, Memphis -73, Long Beach -58, Cleveland -45, Boston -57, Syracuse -48). The average annual net change by spoke was +0.5 veterans and ranged from -49 in Dallas to +79 in Dallas. Six facilities had significant net change (outside ± 2 SD: Aurora +179, Altoona +40, Oklahoma City +37, Dallas +37, Long beach -62, Albany -25). Five facilities had significant change when evaluated as a percent (outside 2 SD: Grand Junction +59%, Beckley 36%, Martinsburg 36%, and Aurora +34%).

Veteran proximity to care was also modeled over digital road network and quantified in miles. This commute metric for all veterans was summarized by facility territory and then analyzed for change over time. For the hub facilities, the average veteran commute was 128 miles, ranging from San Juan at 36 miles to Seattle/Alaska with 266 miles. The five year average veteran commute by spoke territory averaged 46 miles with a minimum of 10 and max of 182. The 5 year net change in commute for the average Hub was 0 miles, and a total net change of +1 miles for the entire country. Facilities with the highest net change were Seattle -12, Aurora -34, San Antonio -7, Memphis +7). Percentage change in average commute values by hub were all between -9 and 4%. The average net change in veteran commute for the spokes was -1 miles and ranged from -34 miles in Sheridan to +18 in Anchorage. Six facilities had significant commute change when evaluated as a percent (outside 2 SD: San Antonio +25%, Cheyenne +23%, Grand Junction -21%, Sheridan -25%, Lincoln -32%, and Coatesville -22%).

Lastly, a simple linear regression analysis was done between the 5 year net change in veteran population and the 5 year net change in average veteran commute. For the hubs and spokes there was no significant relationship ($R^2=0$). A pairwise comparison was done between the facilities that saw significant change in population and also a significant change in average veteran commute (significance measured as being greater than ± 2 SD). Five hubs saw significant change in both categories over the five year period: Memphis, Aurora, Long Beach, and Syracuse; but no spoke facilities.

Methodology:

For the purpose of this project, we split the study population into two groups, by diagnosis category: All Diagnoses and Spinal Cord Injury (SCI) only. Then, the study population was split into two groups, defined by their SCID Facility Status: Hub and Spoke. Four sub-groups were created:

- All Diagnoses by Hub
- All Diagnoses by Spoke
- SCI Only by Hub
- SCI Only by Spoke

For each of these four sub-groups, we performed three major analyses:

- Summaries Over Time and Net Change Over Time on Veteran Population
- Summaries Over Time and Net Change Over Time on Veteran Proximity
- We explored any potential relationships between these two variables.

This analysis was repeated 4 times. Lastly, as requested, we split these deliverables into two separate documents: One focused on the All diagnoses cohort and the second one focused on the entire SCI Only cohort.

The analysis we did on veteran populations relative to each facility consisted of two parts, or chapters. The first was to take the 25 SCI Centers, or hubs, and establish for each a geographic area, a measurement of the veteran population. This was done by creating specially designed Thessian polygons for the facilities of interest that accounted for both spatial proximity as well as human commute by road (ESRI, ArcGIS Pro Desktop V3.3) We then summarized the annual veteran population in proximity to each facility for fiscal years 2019 to 2023 by geocoding veteran addresses and running the Closest Facility spatial analysis from the Network Analyst Toolset (ESRI, ArcGIS Pro Desktop V3.3). We then explored change over time in multiple ways: annual change over time, 5 year net change, 5 year average change, annual percentage change, and 5 year percentage change. To determine significant change, we also expressed net change values as standard deviations. We summarized these values as tables of results, summary statistics tables, and various visual aids such as box plots, histograms, bar charts, and map graphics.

We used the output from the Closest Facility analysis that quantified veteran proximity to their closest facility, and summarize these commute metrics for each facility's territory. Then, similar to the population analysis, we looked at trends over time for these measures, unique to each territory. We then explored change over time in multiple ways: annual change over time, 5 year net change, 5 year average change, annual percentage change, and 5 year percentage change. To determine significant change, we also expressed net change values as standard deviations. We summarized these values as tables of results, summary statistics tables, and various visual aids such as box plots, histograms, bar charts, and map graphics. The next step was to quantify veteran proximity to their closest facility, and summarize these commute metrics for each facility's territory.

These methods were repeated with the same population data but different study area to include the 122 spokes.

Research Questions Answered:

Research Question 1: Total number of Veterans in that catchment area (all those where their closest hub is in that catchment area).

The 5 year average veteran population by Hub was 991 veterans, and ranged from 253 in Hampton to 2,257 in Augusta.

Table 1: Summary of Veteran Population by Hub between FY19 and FY23

City	FY19	FY20	FY21	FY22	FY23
Albuquerque	346	339	338	312	320
Augusta	2183	2231	2257	2217	2242
West Roxbury	1146	1167	1133	1096	1089
Bronx	499	495	460	469	482
Syracuse	704	700	693	681	656
Cleveland	1939	1939	1935	1736	1894
Miami	632	616	615	605	615
Dallas	1656	1677	1667	1627	1728
Aurora	877	939	977	1024	1114
East Orange	690	704	697	676	712
Hines	1146	1140	1130	1048	1160
Houston	588	580	595	566	587
Hampton	257	255	263	257	253
Long Beach	1366	1353	1341	1297	1308
Memphis	1170	1167	1155	1064	1097
Minneapolis	1190	1215	1216	1153	1203
Palo Alto	885	897	925	915	909
Richmond	1501	1523	1532	1571	1626
St Louis	1291	1296	1300	1298	1323
Seattle	924	936	961	905	934
San Diego	531	539	543	555	552
San Antonio	487	487	498	469	512

San Juan	364	363	356	347	345
Tampa	1974	1948	1949	1881	1956
Milwaukee	421	424	417	400	384
Grand Total	24,917	25,085	25,105	24,321	25,157

Research Question 2: Total number of Veterans in each closest-hub-or-spoke for the catchment area (hub and each spoke numbers).

Due to the large amount of data produced in results for question 2, a summary table of 147 columns or rows wasn't added to this report, but is located in the Excel Workbook [GIS Veteran Population Analysis Calculations NO PHI.xlsx](#) in tab 'Spokes_AllDx_'.

A summary table of descriptive statistics on the Veteran Population by spoke territory and fiscal year is given below. The average Spoke population was 170 veterans, and ranged from 25 in Hawaii to 824 in Dallas.

Table 2. Veteran Population by Facility Territory and Fiscal Year					
	FY19	FY20	FY21	FY22	FY23
Mean	170	171	171	165	171
Median	136	133	134	126	135
Mode	76	53	65	119	65
Standard Deviation	124	125	126	124	130
Kurtosis	5	5	5	5	6
Skewness	2	2	2	2	2
Range	786	800	793	744	824
Minimum	25	24	25	25	24
Maximum	811	824	818	769	848
Sum	24,917	25,085	25,105	24,321	25,157
Count	147	147	147	147	147

Research Question 3: Did the population for the entire catchment area, and for each individual spoke and the hub, have a statistically significant increase or decrease during FY19 to FY23?

Hubs: Analysis of table 3 below that there was a large drop in FY22, followed by an equal rise in FY23. The average annual change per facility, per year, was a population gain of 60 veterans. The average change ranged from a loss of 18 veterans per year (Memphis), to an average gain of 59 veterans per year (Aurora). Statistical significance is further addressed with the next question.

Table 3: Summary of Net Change in Veteran Population by Hub from FY19 to FY23

City	FY20	FY21	FY22	FY23	Annual Average	Net Change
Albuquerque	-7	-1	-26	8	-7	-26
AUGUSTA	48	26	-40	25	15	59
Boston	21	-34	-37	-7	-14	-57
Bronx	-4	-35	9	13	-4	-17
SYRACUSE	-4	-7	-12	-25	-12	-48
CLEVELAND	0	-4	-199	158	-11	-45
MIAMI	-16	-1	-10	10	-4	-17
DALLAS	21	-10	-40	101	18	72
AURORA	62	38	47	90	59	237
EAST ORANGE	14	-7	-21	36	6	22
HINES	-6	-10	-82	112	4	14
HOUSTON	-8	15	-29	21	0	-1
HAMPTON	-2	8	-6	-4	-1	-4
LONG BEACH	-13	-12	-44	11	-15	-58
MEMPHIS	-3	-12	-91	33	-18	-73
MINNEAPOLIS	25	1	-63	50	3	13
PALO ALTO	12	28	-10	-6	6	24
RICHMOND	22	9	39	55	31	125
ST LOUIS	5	4	-2	25	8	32
SEATTLE	12	25	-56	29	3	10

San Diego	8	4	12	-3	5	21
SAN ANTONIO	0	11	-29	43	6	25
SAN JUAN	-1	-7	-9	-2	-5	-19
TAMPA	-26	1	-68	75	-5	-18
MILWAUKEE	3	-7	-17	-16	-9	-37
Grand Total	168	20	-784	836	60	240

Research question 4: Are there catchment areas, or individual hubs and spokes, which have had the most increase or decrease in Veterans living closest? Another way to ask the question is, are there any geographic catchment areas, or individual hubs and spokes, that are significantly growing or shrinking?

Hubs: to answer question 4, we extracted data from table 3 highlighting the top 5 largest net population change increases and decreases over the last five fiscal years:

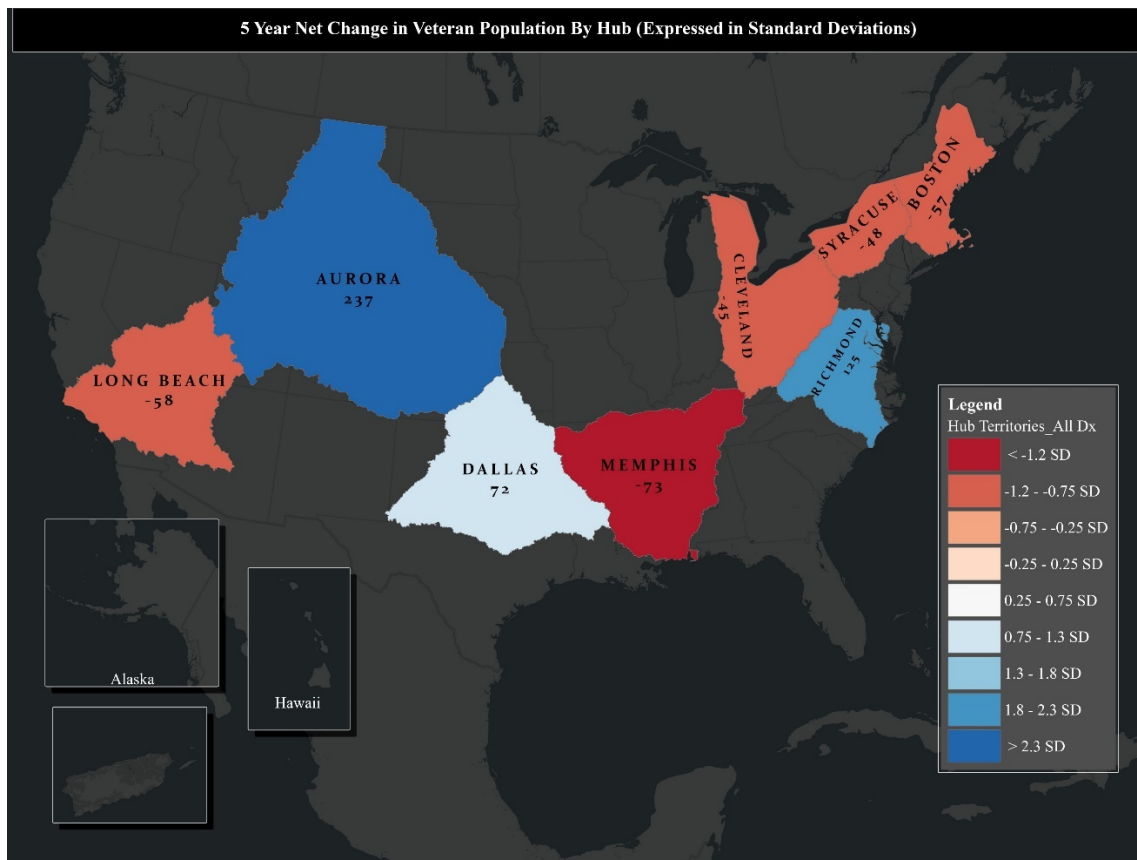
Top 5 net population change (decline) between FY19-23.

1. Memphis, -73
2. Long Beach, -58
3. West Roxbury, -57
4. Syracuse, -48
5. Cleveland, -45

Top 5 net population change (increase) between FY19-23:

1. Aurora, 237
2. Richmond, 125
3. Dallas, 72
4. Augusta, 59
5. St. Louis, 32

We also examined where the most extreme change was for the five year period. We took the net change values expressed as standard deviations, and filtered the map graphic to only show those values that were 1 or more standard deviations away from the mean, so non of the middle values are shown. In this way we objectively delineated which facilities had the *most* significant change relative to the whole. Richmond, Dallas, and Aurora had the most significant 5 year gains, and Long Beach, Memphis, Cleveland, Syracuse, and Boston had the most significant five year losses.



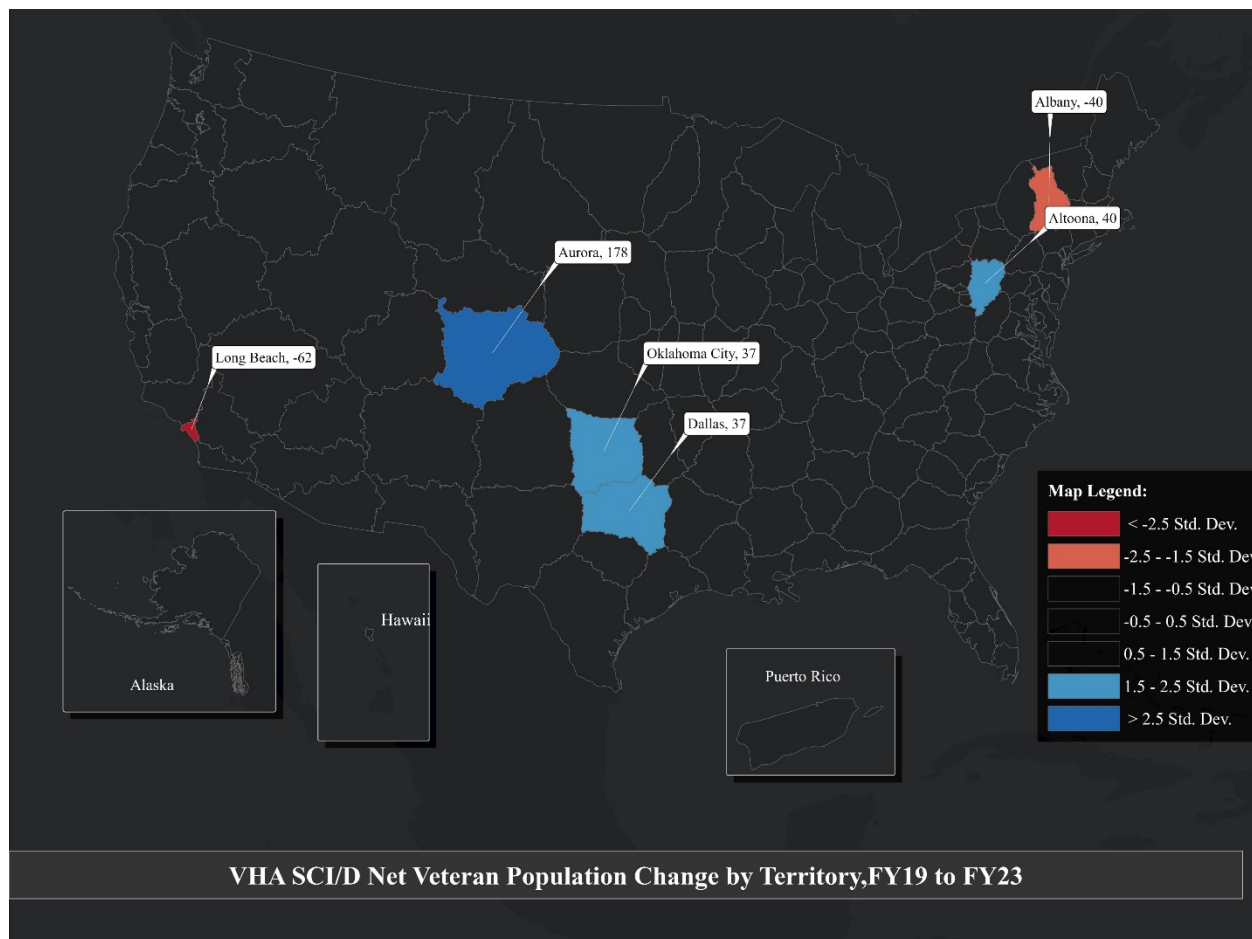
Spokes: Due to the large amount of data produced in results for question 4, a summary table of 147 columns or rows wasn't added to this report, but is located in the Excel Workbook [GIS Veteran Population Analysis Calculations NO PHI.xlsx](#) in tab 'Spokes_AllDx_'. A summary table of descriptive statistics on the Net Change in Veteran Population by spoke territory and fiscal year is given below in Table 4.

Grand Summary Table Of Net Change In Veteran Population By Fiscal Year for Spokes

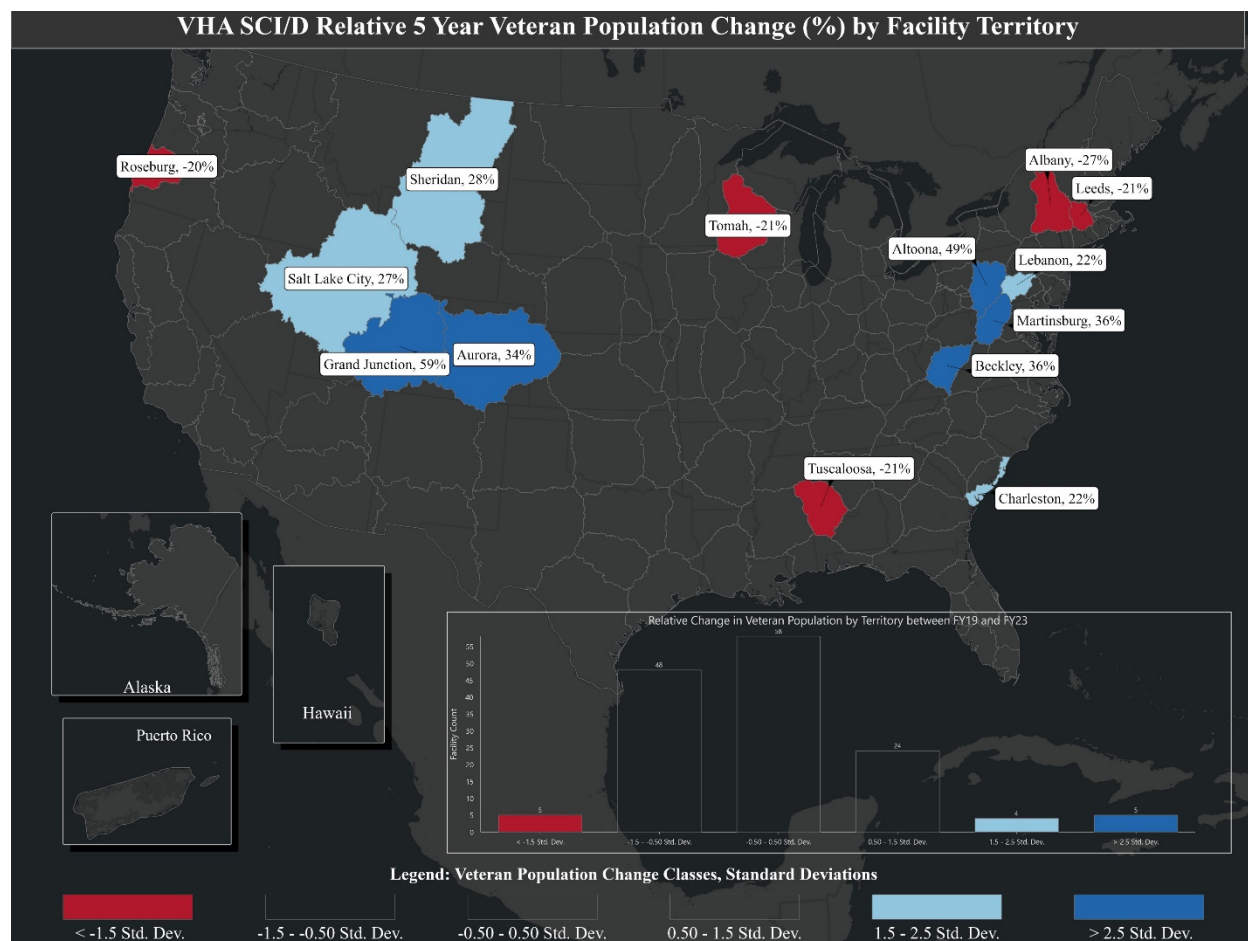
	FY20	FY21	FY22	FY23	5Y
Mean	1	0	-5	6	2
Median	1	-1	-3	2	0
Mode	2	1	-7	-2	-6
Standard Deviation	7	7	12	14	21
Kurtosis	16	5	3	9	32
Skewness	2	1	-1	3	4
Range	72	54	76	91	240
Minimum	-18	-17	-49	-12	-62
Maximum	54	37	27	79	178
Sum	168	20	-784	836	240
Count	147	147	147	147	147

The average net change across all 147 facilities for any given year is single digits. The minimum net change in territory veteran population ranged from a loss of 12 veterans in FY23 to a loss of 49 veterans in FY22. The largest population gain for a territory was 27 in FY22 and 79 in FY23. The sum row shows the total net gain or loss for all territories together. The count row is the number of facility territories included in the analysis. The standard deviations are much smaller than the hubs, and the range is also much smaller. The range in territory population change was 54 veterans in FY21, and 91 in FY23.

Lastly, The five year change column looks at the difference between FY23 and FY19 without considering the middle three years. These numbers are larger because they are taking into account 5 years of cumulative change, while all the other columns are looking at change over a one year period.



Six spoke facilities had significant net change (outside ± 2 SD: Aurora +178, Altoona +40, Oklahoma City +37, Dallas +37, Long beach -62, Albany -25).



Five spoke facilities had significant change when evaluated as a percent (outside 2 SD: Grand Junction +59%, Beckley 36%, Martinsburg 36%, and Aurora +34%).

Research Question 5: For each catchment area, and each individual hub and spoke, provide the average drive time and distance to closest hub, and closest hub-or-spoke.

Table 5 (below), summarizes the average veteran commute by their hub territory and by fiscal year and the units are in miles. Each column in the table represents a fiscal year and each row is for a specific territory. So as an example interpretation of the table, at the intersection of column FY19 and row Albuquerque this table says “In the Albuquerque hub territory in FY19, the average commute distance for all veterans was 131 miles.”

There is a lot of variation in the summary results for each column, which makes sense when we remember the orange map at the beginning of the presentation that showed the large variation of geographic area that each hub facility had. However when you look at a specific row you'll notice that there is a very little variation over the years in the average commute value for any given territory.

Table 5: Summary of Average Veteran Commute (miles) by Hub and Fiscal year

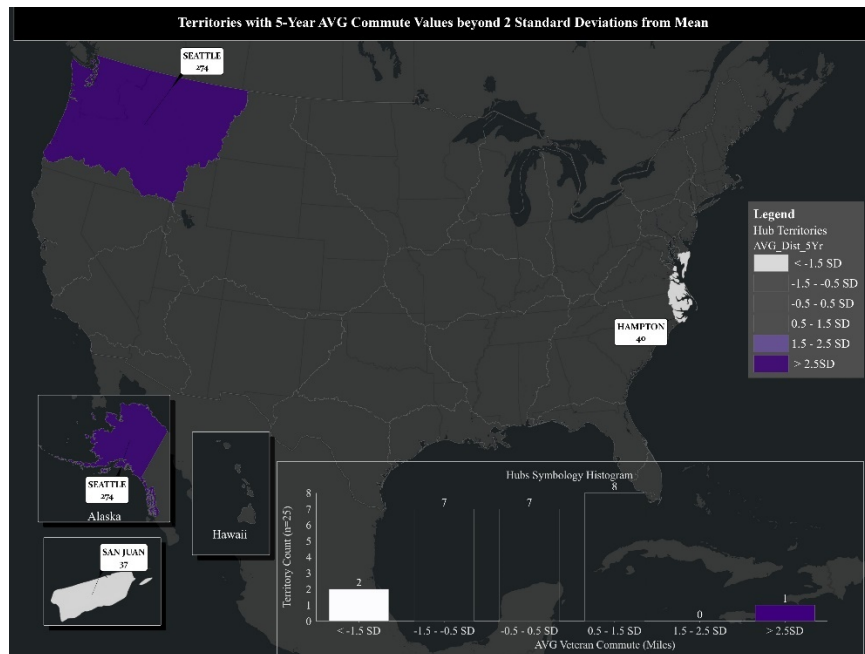
City	FY19	FY20	FY21	FY22	FY23
Albuquerque	131	132	134	135	133
AUGUSTA	173	173	173	173	173
WEST ROXBURY	96	95	96	98	98
Bronx	45	46	46	45	45
SYRACUSE	97	96	95	94	92
CLEVELAND	165	167	166	168	167
MIAMI	47	47	47	46	46
DALLAS	144	145	144	149	144
AURORA	183	176	172	175	172
EAST ORANGE	91	91	94	89	95
HINES	110	110	112	109	113
HOUSTON	123	126	128	124	125
HAMPTON	38	41	41	39	40
LONG BEACH	177	177	178	181	181
MEMPHIS	202	202	205	205	209
MINNEAPOLIS	171	168	168	172	172
PALO ALTO	136	134	134	134	134
RICHMOND	140	139	140	140	141
ST LOUIS	186	185	183	185	186
SEATTLE	278	275	279	271	266
San Diego	187	184	187	184	184
SAN ANTONIO	83	83	79	79	77
SAN JUAN	37	36	36	37	37
TAMPA	110	110	113	113	114
MILWAUKEE	79	80	78	79	77
Grand Average:	143.6	143.5	144.3	144.3	144.4

Spokes:

Table 6 Summary of Average Veteran Commute to Spoke by Fiscal Year										
	Min	Max	Mean	S.D.	Media n	Count	Range	IQR	Q1	Q3
FY19	10	182	46	25	41	147	171	25	30	55
FY20	11	159	46	24	41	147	148	24	29	54
FY21	11	176	46	24	41	147	165	24	30	54
FY22	11	161	46	24	41	147	150	25	30	55
FY23	10	174	45	24	41	147	164	24	30	54

The five year average veteran commute by spoke territory averaged 46 miles with a minimum of 10 and max of 182. The average net change in veteran commute for the spokes was -1 miles and ranged from -34 miles in Sheridan to +18 in Anchorage. Six facilities had significant commute change when evaluated as a percent (outside 2 SD: San Antonio +25%, Cheyenne +23%, Grand Junction -21%, Sheridan -25%, Lincoln -32%, and Coatesville -22%).

Research Question 5a: Do certain hubs, certain spokes, and certain overall catchment areas, have Veterans that have to drive statistically significantly further, or spend more time driving, than other catchment areas, and does this change over time (FY19-23)?



San Juan had had the lowest 5 year average of the average veteran commute values at 37 miles, while Hampton was close behind with 40 miles for its 5 year average veteran commute. Seattle had the largest 5 year average, average veteran commute at 274 miles.

To examine net change, we extracted data from table 4 highlighting the top 5 largest net increase and decrease in commute distance over the last five fiscal years:

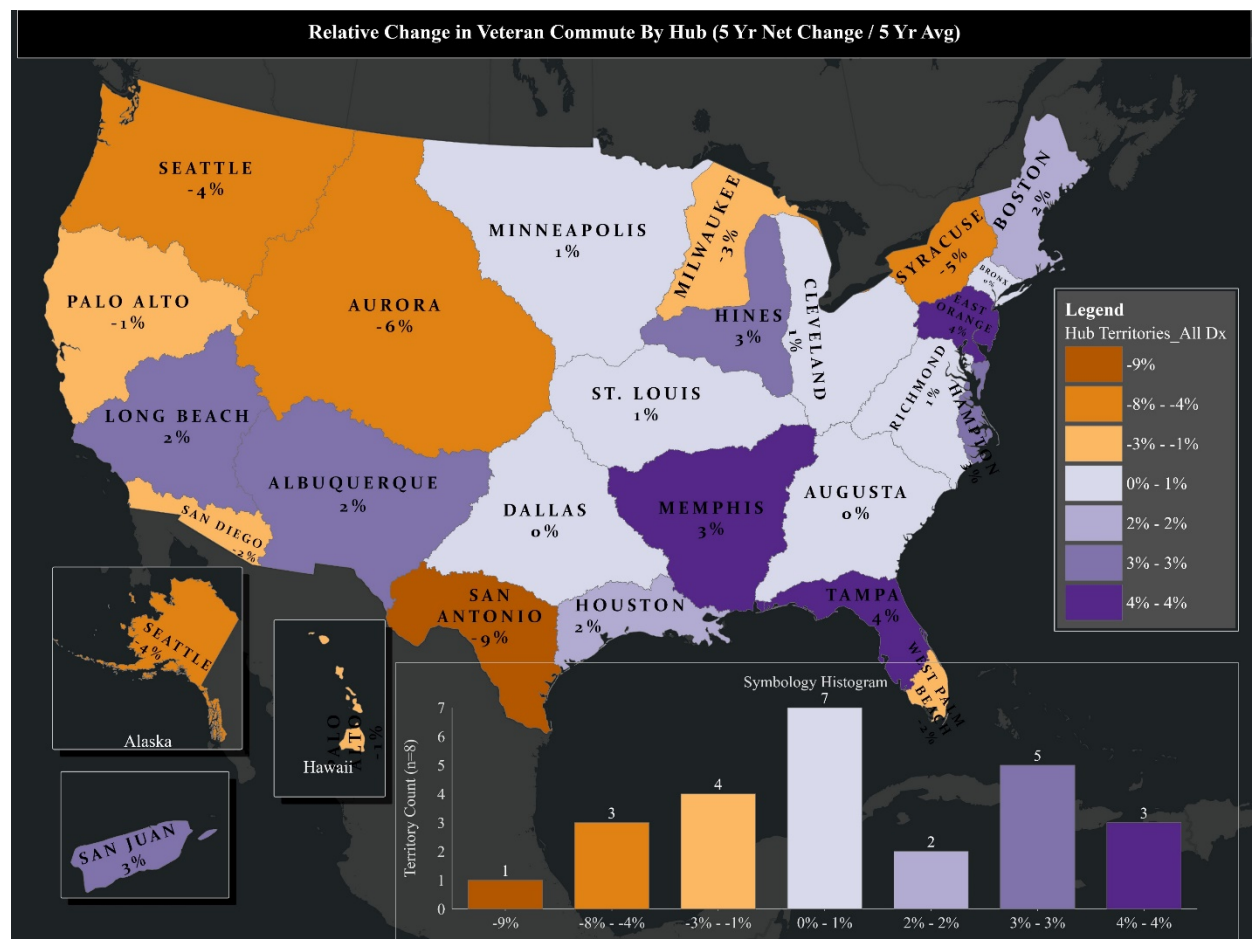
Top 5 Net change – Vet population with increase in commute distance (FY19-23):

1. Memphis, 7
2. Tampa, 4
3. Long Beach, 4
4. East Orange, 4
5. Hines, 3

Top 5 Net change – Vet population with decrease in commute distance (FY19-23):

1. Seattle, -12
2. Aurora, -11
3. San Antonio, -7
4. Syracuse, -5
5. San Diego, -4

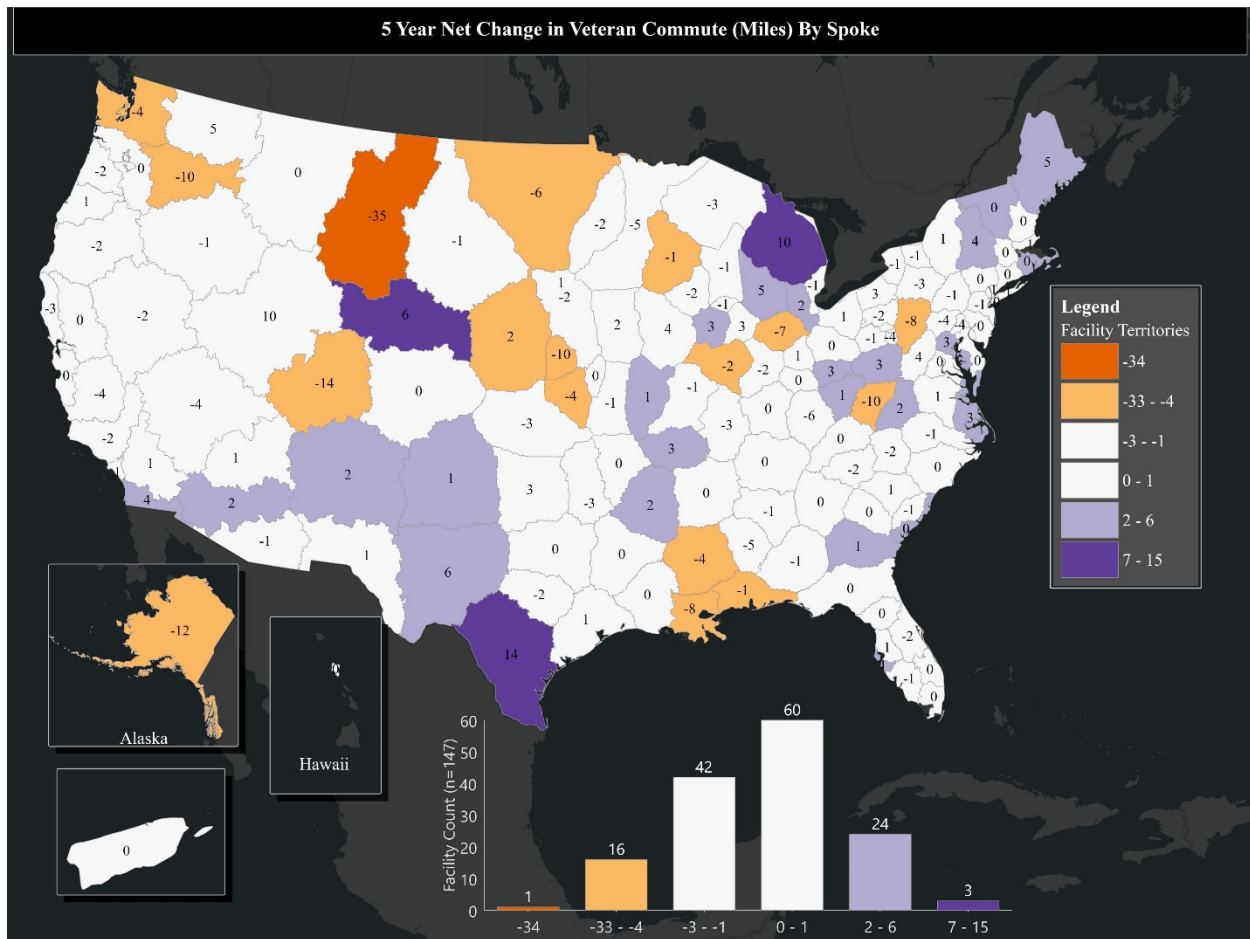
Map Graphic 3: Relative change in Veteran Commute by Hub (5 year net change/5 year average)
 Lastly, we looked at relative change in commutes by normalizing net change by the average value.



We examined the relative change in commute that's proportional to the average commute value for each territory. This takes into account the size of each territory, as well as giving us insight into how big of a change each territory saw relative to itself.

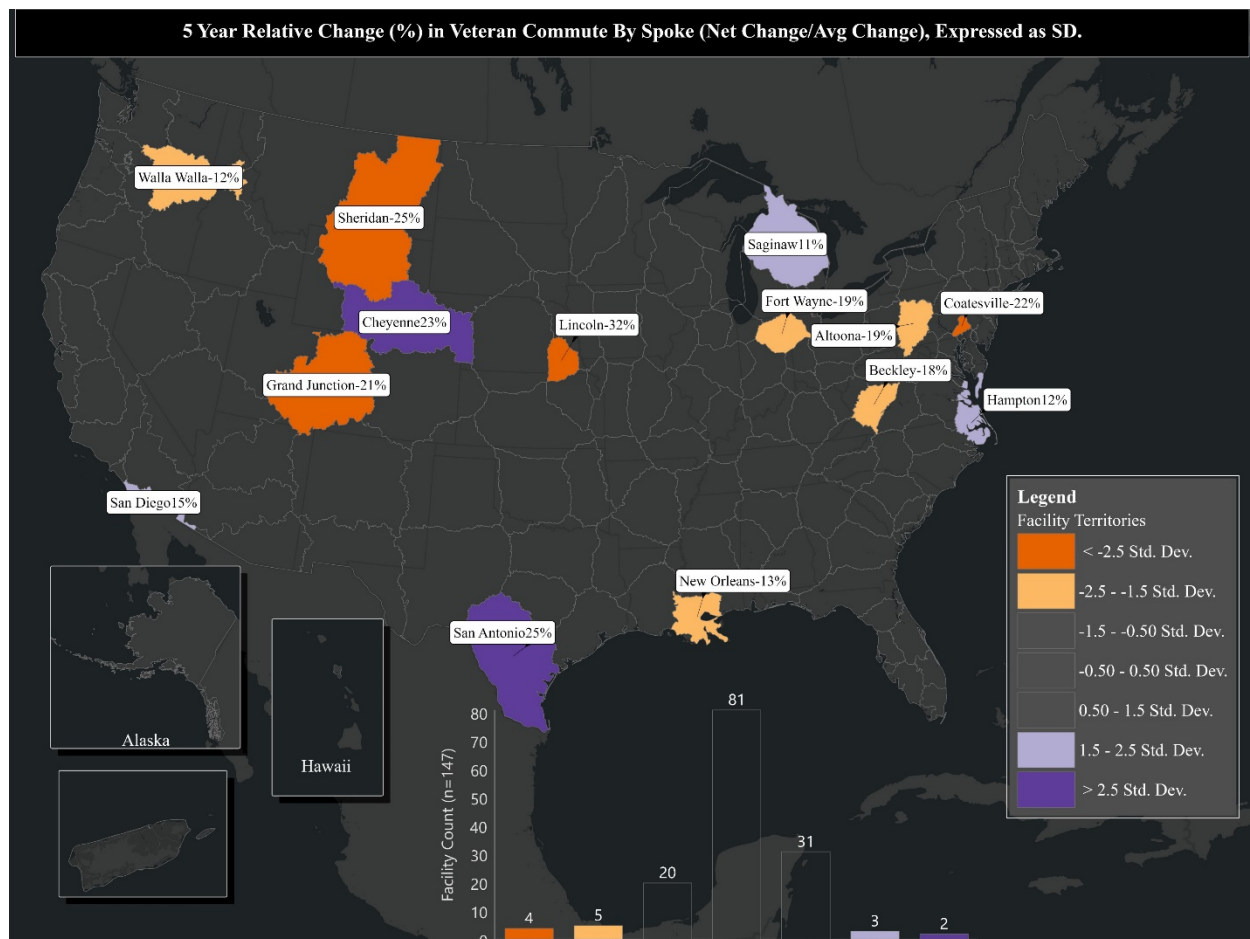
If we normalize the absolute change in veteran commute for each territory by the average commute for each territory, we get to see change values proportional to each territory. If you look at either the legend or the histogram of map graphic 3 (below), the color ramp is the same. Percentage change in average commute values by hub were all between -9 and 4%. So while the color symbology covers an extreme range of dark and light diverging hues that seems to suggest significant relative change, the data itself tempers that message. If we were to stretch the color ramp from -100 to +100 percent change, all 25 values would be in the pale purple and orange classes.

Spokes:



This graphic summarizes the 5 year net change in veteran commute by spoke facility. The average net change in veteran commute for the spokes was -1 miles and ranged from -34 miles in Sheridan to +18 in Anchorage.

We also analyzed relative change in commutes for the spokes that's proportional to the average commute value for each territory. This takes into account the size of each territory, as well as giving us insight into how big of a change each territory saw relative to itself. If we normalize the absolute change in veteran commute for each territory by the average commute for each territory, we get to see change values proportional to each territory. We filtered out the middle range of values, we can again use standard deviations to classify the data into 7 classes.



This graphic summarizes where the greatest 5 Year relative change was in patient commutes by visualizing only those data values that were two standard deviations or greater away from the mean. The greatest relative change centered around the Rockies and San Antonio. Six facilities had significant commute change when evaluated as a percent (outside 2 SD: San Antonio +25%, Cheyenne +23%, Grand Junction -21%, Sheridan -25%, Lincoln -32%, and Coatesville -22%).

Research Question 5b: Is there a relationship between catchment area or hub population size and change FY19-23, and drive time/distance? In other words, do catchment areas who have a relative, significant population increase over time, have less average drive time or distance, and would the reverse be true for catchment areas and hubs with relative, significant population decreases?

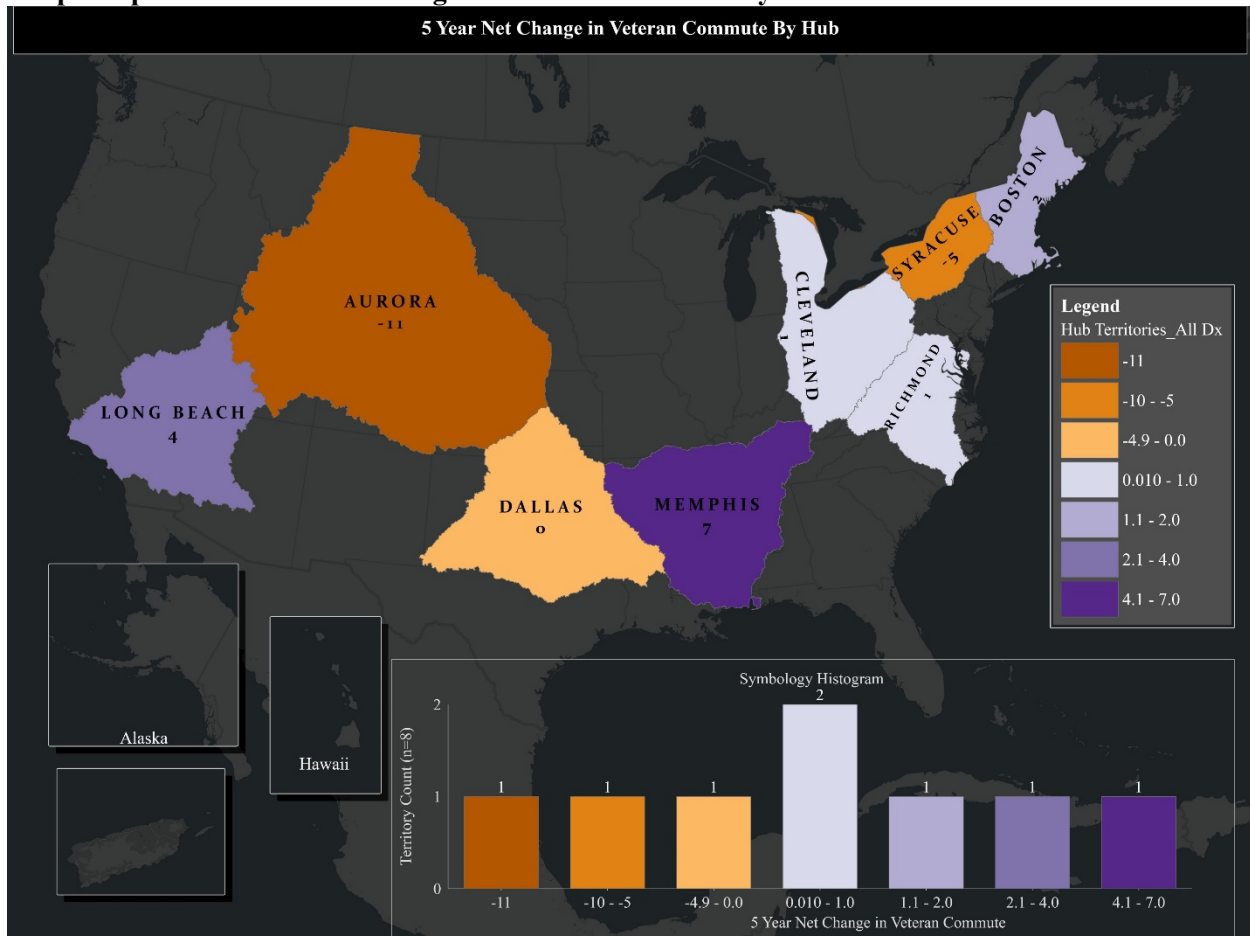
To address question 5b, a simple linear regression analysis was done between the 5 year net change in veteran population and the 5 year net change in average veteran commute. For the hubs and spokes there was no significant relationship ($R^2=0$).

A pairwise comparison was also done between the facilities for significant change in population and also a significant change in average veteran commute (significance measured as being greater than ± 2 SD).

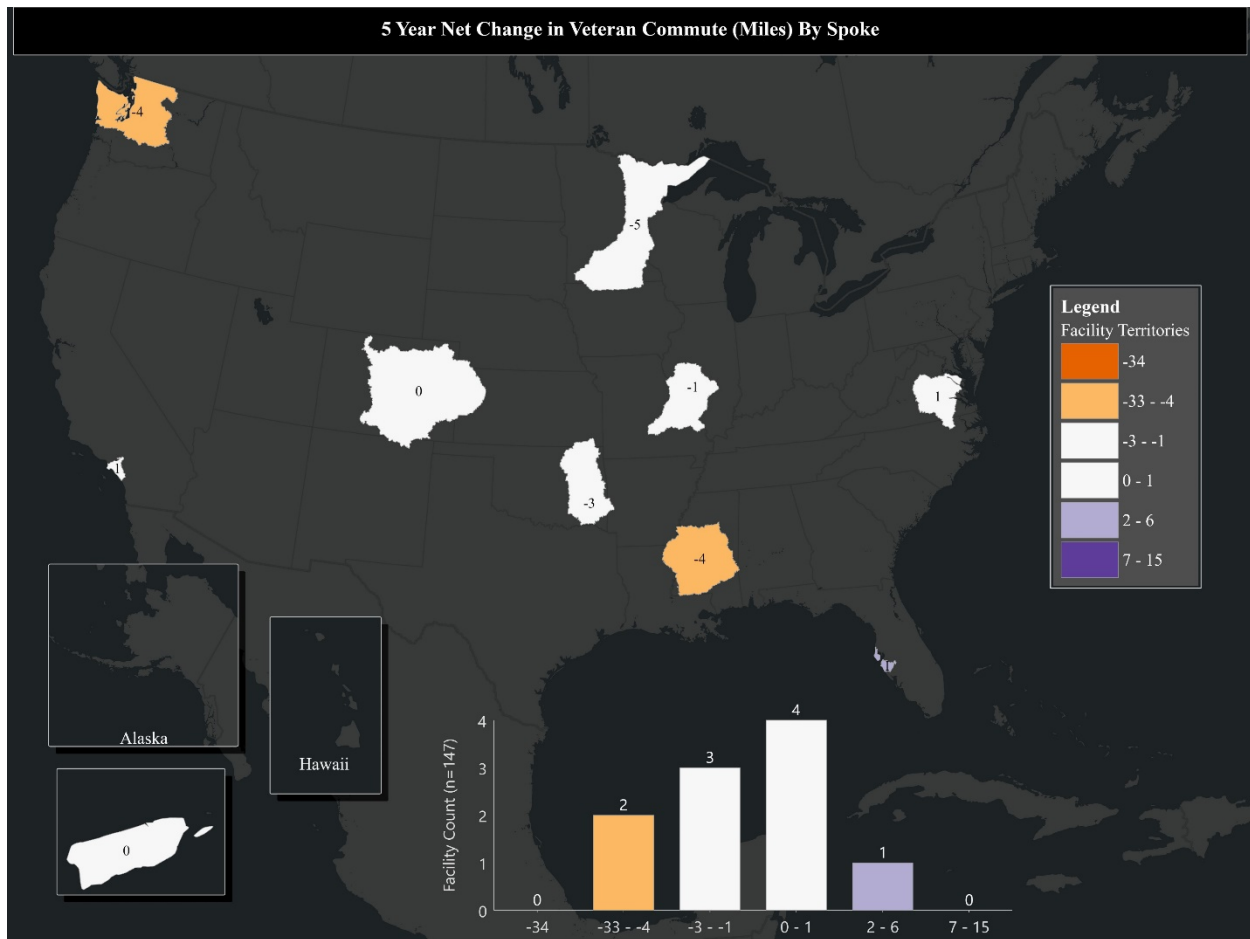
The map graphic from the population change section was used as a filter by the territories that only showed the Hubs that had the highest range of change in population, or the facilities with the highest

gain and lowest losses. Then we applied that filter on the net change in commute values, and filtered those 25 territories by the 8 territories that had significant change from the population change map.

Map Graphic 6: 5 Year net Change In Veteran Commute By Hub



Five hubs saw significant change in both categories over the five year period: Memphis, Aurora, Long Beach, and Syracuse.



The pairwise comparison for the spoke facilities' results in significant change values did not have any matches, as shown in the image above. In both the legend and the symbology histogram, these net change in commute values were not in any of the most significant ranges of the data. This further reinforces the results of the SLR model that there was little relationship between the net change in veteran population by spoke, and net change in veteran commute by spoke.